

Unsupervised Learning: Implementation of K-Means Clustering Algorithms

1. Objective

The objective of this task is to understand the concept of unsupervised learning by implementing the K-Means clustering algorithm. The task aims to group similar customers based on their Annual Income and Spending Score without using labeled data. Additionally, a comparison of different clustering algorithms is provided to understand their characteristics and applications.

2. Tools & Libraries Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

3. Dataset Information

Dataset Name:

Mall Customers Dataset

Number of Records:

200

Number of Features:

5

Attributes:

Column	Description
CustomerID	Unique ID of each customer
Genre	Gender of the customer
Age	Customer's age
Annual Income (k\$)	Annual income in thousand dollars
Spending Score (1-100)	Spending score assigned by the mall

4. Methodology

Step 1

Imported the required Python libraries.

Step 2

Loaded the Mall Customers dataset into Google Colab.

Step 3

Checked the dataset structure and verified that there were no missing values.

Step 4

Selected **Annual Income (k\$)** and **Spending Score (1-100)** as the features for clustering.

Step 5

Visualized the customer distribution using a scatter plot.

Step 6

Applied the Elbow Method to determine the optimal number of clusters.

Step 7

Implemented the K-Means clustering algorithm with **K = 5**.

Step 8

Visualized the clustered customers along with their cluster centroids.

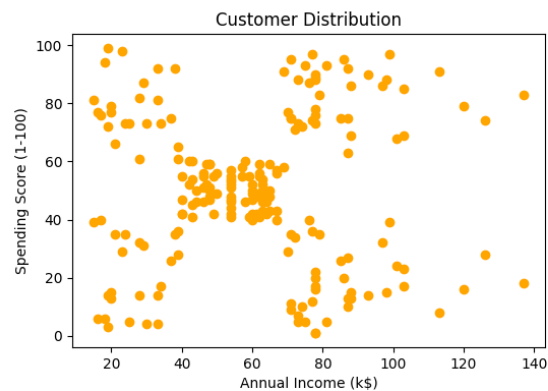
Step 9

Saved the clustered dataset as a CSV file.

5. Results

Graph 1

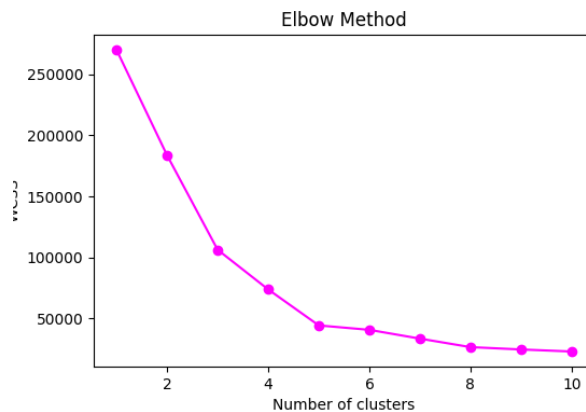
Customer Distribution



This scatter plot shows the distribution of customers based on Annual Income and Spending Score before clustering.

Graph 2

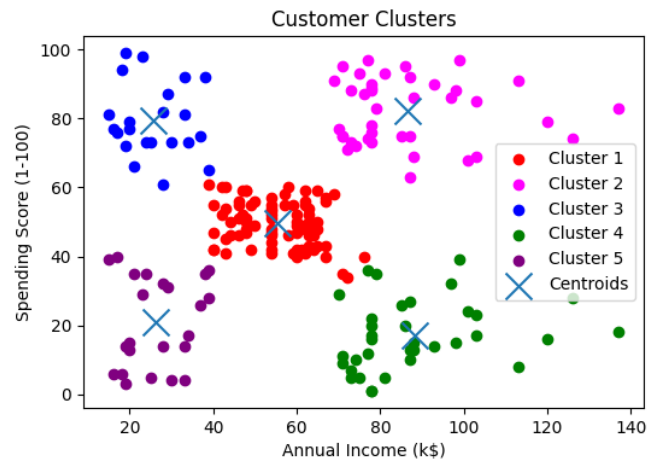
Elbow Method



The Elbow Method was used to determine the optimal number of clusters. The graph indicates that **K = 5** is an appropriate choice.

Graph 3

K-Means Clustering



The K-Means algorithm grouped customers into five distinct clusters based on their Annual Income and Spending Score. The centroids represent the center of each cluster.

6. Comparison of Clustering Algorithms

Feature	K-Means	Hierarchical Clustering	DBSCAN
Type	Centroid-based	Hierarchical	Density-based
Number of Clusters	User-defined (K)	Determined using dendrogram	Automatically detected
Speed	Fast	Slower	Moderate
Handles Noise	No	Limited	Yes
Best For	Large datasets	Small to medium datasets	Arbitrary-shaped clusters
Main Limitation	Requires choosing K	High computational cost	Sensitive to parameter selection

7. Conclusion

In this task, the K-Means clustering algorithm was successfully implemented on the Mall Customers dataset. The Elbow Method was used to determine the optimal number of clusters, and customers were grouped into five meaningful clusters based on Annual Income and Spending Score. K-Means proved to be an efficient algorithm for customer segmentation. A comparison with Hierarchical Clustering and DBSCAN highlighted the strengths and limitations of different clustering approaches.